Interactive Natural Selection activity **Name** _____

http://projectbeak.org/adaptations/build.htm - play the game

1. What is the highest percentage chance of survival you achieved?
   _____

2. Some birds don't survive well in some habitats because they can't get enough food. List three other ways a bird's survival could be affected in the environment.

   _____

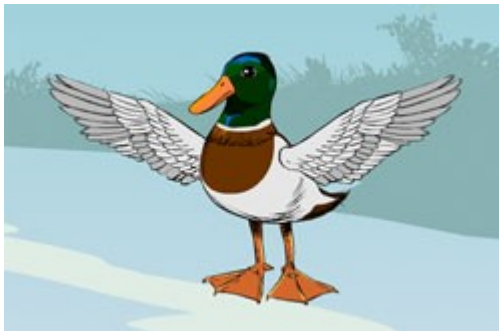
   _____

   _____

3. If you made a bird with a high chance of survival (more than 70%), would you expect that type of bird to produce more or less chicks than other birds?

   _____

4. Create the bird pictured below in the pond habitat and record its chance of surviving. Make the same bird in two other habitats and record its chances of surviving.



Habitat	Chance of surviving
Pond	

5. What happened when you changed the environment of your bird? Was it better or more poorly adapted to its new environment?

   _____

   _____

   _____

6. If the habitat of this type of bird changed from dead trees to ponds, would you expect to see more or less of them in the next generation? Why?

   _____

   _____

   _____

   _____

http://peppermoths.weebly.com/

Read Chapter 1 – Life cycle of the peppered moth.

1. What animals prey on the peppered moth? _____
2. Describe three colour variations in peppered moths.
   _____
   _____
   _____

Read Chapter 2 - Pollution and peppered moths.

3. When was the first black form of a peppered moth recorded?
   _____
4. What caused moth populations to change colour during the industrial revolution?
   _____
   _____
   _____

Read Chapter 3 – Dr Kettlewell tests natural selection.

Do the simulation "A birds eye view of natural selection" and fill in the table below:

Types of moths	**Percentage of moths**	
	Unpolluted forest (light coloured trees)	Polluted forest (dark coloured trees)
Light moths		
Dark moths		

5. In unpolluted forests, which moths survived better? _____
6. Which moths were better suited to the polluted forests?
   _____
7. Find the meaning of the term 'natural selection' and explain its meaning, using either the peppered moth or the beetle as an example. Use the words 'survive' and 'reproduce' in your explanation.